Separable Differential Equations

ANSWER KEY

Solving by separation of variables

1 Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
$y\,dy = x\,dx$, so $\frac{y^2}{2} = \frac{x^2}{2} + C$. Using $y(0) = 3$: $C = \frac{9}{2}$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (taking the positive root since $y(0) = 3 > 0$).

2 Solve $\frac{dy}{dx} = 2xy$, given $y(0) = 5$.
$\frac{1}{y}dy = 2x\,dx$, so $\ln|y| = x^2 + C$. Using $y(0) = 5$: $C = \ln 5$. Solution: $y = 5e^{x^2}$.

Growth, decay, and applications

3 A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.

a Solve for $P(t)$. $P(t) = 2000e^{0.05t}$.

b Find $P(20)$, to the nearest whole number. $P(20) = 2000e^{1} \approx 5437$.

4 Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a $20°$C room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.
Solving: $T(t) = 20 + 80e^{-kt}$. Using $T(5) = 60$: $40 = 80e^{-5k}$, so $e^{-5k} = \frac{1}{2}$, giving $k = \frac{\ln 2}{5} \approx 0.1386$. So $T(t) = 20 + 80e^{-0.1386t}$.